

ZHE XUE

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EDUCATION

Southern University of Science and Technology, China

Sep. 2023 – Jun. 2027

B.Eng. candidate in Computer Science and Technology (Turing Class)

- **GPA:** 3.92/4.00; **Rank:** 4/157
- **Peer Tutor, Linear Algebra** (Spring & Fall 2024): delivered exam review sessions for fellow students.

AWARDS

Gold Medal, 2024 ICPC Asia Chengdu Regional Contest

Oct. 2024

Gold Medal, 2024 ICPC Asia Hong Kong Regional Contest

Dec. 2024

Silver Medal, 2024 ICPC Asia East Continent Final Contest

Dec. 2024

RESEARCH EXPERIENCE

Hybrid Sampling for Reliable On-Policy Distillation

May 2026 – Jul. 2026

- Developed a likelihood-ratio-guided hybrid sampling rule to selectively replace student-proposed tokens with teacher samples before prefix deviations degrade subsequent teacher supervision.
- Formulated a source-aware dual-divergence objective using reverse KL for retained student tokens and forward KL for teacher corrections, without auxiliary reward or verifier models.
- Built a hybrid rollout pipeline on DAPO-Math-17K by adapting vLLM's speculative-decoding engine; end-to-end training incurred 8.6% runtime overhead over OPD under a matched training-data budget.
- Distilled Qwen3-4B into Qwen3-1.7B, improving macro Avg@8 by 3.25 pp and macro Pass@8 by 4.94 pp over OPD on AIME 2024/2025 and AMC 2023.

Uncertainty-Aware Inference for LLM Reasoning

Sep. 2025 – Present

- Characterized temperature-dependent reasoning across four Qwen3 models and five MATH difficulty levels, tracking answer accuracy, ReasonEval validity/redundancy, and Math-Shepherd verification scores.
- Observed sharp degradation beyond model- and difficulty-dependent temperature thresholds; harder problems degraded at lower temperatures.
- Benchmarked majority voting and confidence-weighted aggregation under temperature sweeps, finding that multi-sample gains peaked before high-temperature degradation.
- Investigating model-internal signals of uncertainty to guide adaptive inference for LLM reasoning.

PROJECTS

SysY Compiler^①

Jul. 2025

- Built a compiler written in Rust for a C-like language, lowering ASTs through Koopa IR to RV32IM assembly.
- Implemented scoped symbol tables, short-circuit control flow, functions, and multidimensional arrays.
- Optimized IR with 7 rewrite rules and code generation with a 13-register write-back cache.
- Passed 130 functional tests in both Koopa IR and RISC-V modes, including 20 benchmark programs.

RISC-V CPU^②

Apr. 2025 – May 2025

- Designed a five-stage pipelined RV32I CPU in Verilog supporting 38 instructions.
- Handled data/control hazards with forwarding, load-use stalls, static branch prediction, and pipeline flushing.
- Implemented UART loading of instruction/data memory and a 3-stage Python deployment toolchain.
- Passed 8 Vivado simulation tests and 3 on-board assembly tests covering core modules, MMIO, and AUIPC.

SKILLS

- **Languages:** Mandarin (Native), English (TOEFL iBT: 5.0/6.0, comparable 100/120)
- **Programming:** C++, Python, Rust, Java, Haskell, CUDA
- **Tools:** PyTorch, vLLM, Docker, Git, Bash, L^AT_EX